

The background of the slide is a close-up, slightly out-of-focus photograph of numerous colorful wooden blocks. The blocks are in various colors including yellow, red, blue, green, pink, and purple, and are arranged in a way that suggests they are being built or dismantled, creating a sense of structure and complexity.

Implementation DEMO

Java Collections Framework Reference

List				
	<code>get(i)</code>	<code>set(i, x)</code>	<code>add(i, x)</code>	<code>remove(i)</code>
<code>ArrayList</code>	$O(1)$	$O(1)$	$O(1 + n - i)$	$O(n - i)$
<code>LinkedList</code>	$O(1 + \min\{i, n - i\})$			

Deque				
	<code>addFirst(x)</code>	<code>removeFirst()</code>	<code>addLast(x)</code>	<code>removeLast(x)</code>
<code>ArrayDeque</code>	$O(1)$			
<code>LinkedList</code>	$O(1)$			

Queue			
	<code>add(x)</code>	<code>remove()</code>	<code>element()</code>
<code>ArrayDeque</code>	$O(1)$		
<code>LinkedList</code>	$O(1)$		
<code>PriorityQueue</code>	$O(\log n)$	$O(\log n)$	$O(1)$

Java Collections Framework Reference

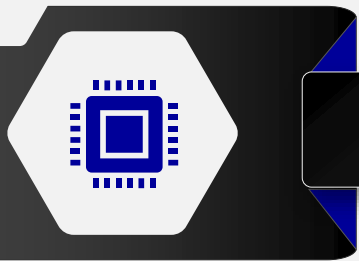
Set			
	<code>add(x)</code>	<code>remove(x)</code>	<code>contains(x)</code>
HashSet	$O(1)$		
TreeSet	$O(\log n)$		

SortedSet			
	<code>headSet(y)</code> ¹	<code>tailSet(x)</code> ¹	<code>subSet(x, y)</code> ¹
TreeSet	$O(\log n)$		

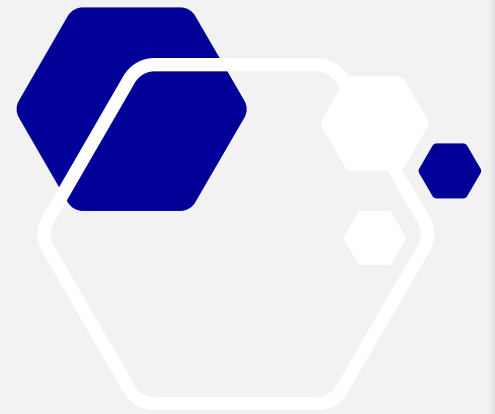
Map ²			
	<code>get(k)</code>	<code>put(k, v)</code>	<code>containsKey(k)</code>
HashMap	$O(1)$		
TreeMap ³	$O(\log n)$		

Java Collections Framework Reference

Collections			
<code>sort(list)</code>	<code>min(c)/max(c)</code>	<code>reverse(list)</code>	<code>binarySearch(list, x)</code> ⁴
$O(n \log n)$	$O(n)$	$O(n)$	$O(\log n)$



ListDemo.java & SetDemo.java



> javac ListDemo.java

← to compile

> java ListDemo

← to run

With sets order is not guaranteed.

Demonstration

> javac ListDemo.java

← to compile

> java ListDemo

← to run

attribute

> java ListDemo 100000000

Adding 100000000 values...done (2.732694899s)

> java ListDemo 100000

Adding 100000 values...done (0.0066799000000000001s)

Adding 100000 values at the front...done (0.322008199s)

> java ListDemo 1000000

Adding 1000000 values...done (0.0279675010000000002s)

Adding 1000000 values at the front...done (65.229541s)

Huge difference!

Now change implementation from ArrayList to LinkedList

> java ListDemo 1000000

Adding 1000000 values...done (0.1105192000000000001s)

Adding 1000000 values at the front...done (0.1201882000000000001s)

> java ListDemo 100000

Using 200000 calls to contains()...done (35.2039442s)

Very slow ☹️

Demonstration

> java SetDemo 100000

Adding 100000 elements to HashSet ...done (0.0240651000000000002s)

Searching 200000 elements...done (0.01108030000000000001s)

Adding 100000 elements to TreeSet ...done (0.0842638s)

Searching 200000 elements...done (0.05290440000000000004s)

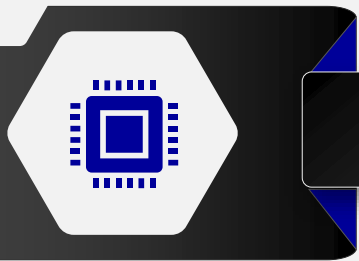
> java SetDemo 10000000

Adding 10000000 elements to HashSet ...done (1.098738s)

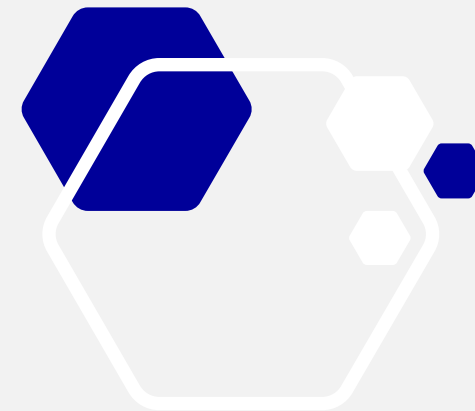
Searching 20000000 elements...done (0.3798946s)

Adding 10000000 elements to TreeSet ...done (2.732916300000000003s)

Searching 20000000 elements...done (2.5870101s)



WordFun.java



This program opens a file and parses it into words.
It returns the content of the file as a giant list of strings.

```
> java WordFun TreasureIsland.txt
```

the book is copyright free so
we can freely use it as test data

shell command to count # of words:

```
> wc -w TreasureIsland.txt
```

```
> cat TreasureIsland.txt | Measure-Object -Word
```

PowerShell

Basic PowerShell Commands

cat	Get-Content	Gets the contents of a file.
cd	Set-Location	Sets the current working location to a specified location.
chdir	Set-Location	Sets the current working location to a specified location.
clc	Clear-Content	Deletes the contents of an item, but does not delete the item.
clear	Clear-Host	Clears the display in the host program.
cp	Copy-Item	Copies an item from one location to another.
ls	Get-ChildItem	Gets the files and folders in a file system drive.
man	help	Displays information about Windows PowerShell commands and concepts.
md	mkdir	Creates a new item.
rm	Remove-Item	Deletes files and folders.